

Quiz #6 Answer Sheet

CS 225 Data Structures

Spring 2026

7:00-8:00pm, Thursday, May 7

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- This is a **closed book** and **closed notes** exam. No electronic aids are allowed, either.
- You should have 4 problems total on 4 pages. You must write on the exam.
- Unless otherwise stated in a problem, assume the best possible design of a particular implementation is being used.
- Unless the problem specifically says otherwise, (1) assume the code compiles, and thus any compiler error is an exam typo (though hopefully there are not any typos), and (2) assume you are NOT allowed to write any helper methods to help solve the problem, nor are you allowed to use additional arrays, lists, or other collection data structures unless we have said you can.
- Please put your name at the top of each page.
- Please fill in your **Lab Section** or your **Lab TA's name** to help us organize and grade your papers.
- Don't panic and good luck!

Problem 1	<u>10</u>	/10
Problem 2	<u>8</u>	/10
Problem 3	<u>7</u>	/10
Problem 4	<u>10</u>	/10
Total	<u>35</u>	/40



Name:

2

Problem 1

1. a) 2 b) 3 (c) A
2. A 3. d 4. b 5. C

Problem 2 Answer sheet

1 C
2 AC
3 BC
4 BD
5 A



Problem 3

A. $\text{pixel.h} / 360.0f$ A₂ pixel.s A₃ pixel.l

B. _____

B1: $\text{convertToPoint}(\text{tiles}[i].\text{avgColor})$ B3: $\text{!pointToIndex}(\text{exist}(p))$ find == endl)B5: tilePoints B6: $\text{convertToPoint}(\text{regions}[i])$ B7: $\text{tree.findNearestNeighbor}(\text{query})$ B8: $\text{pointToIndex}[\text{nearest}]$

C1.

Tile index

0

1

2

3

Point<3>

(0, 1, 0)

(0.5, 1, 0.5)

(0.25, 0.5, 0.5)

(0, 1, 0.5)

C2. $[0, 1, 2, 2]$ C3. $\text{tile } 3 \text{ is duplicate as tile } 0.$ $\text{We use the smallest index here.}$

D.

 $\text{pointToIndex}[p] = (\text{int}) i.$ $\text{if} (\text{pointToIndex.find}(p) == \dots \text{.endl})$ $\text{pointToIndex}[p] = i;$ ~~emplace-back~~B2: $\text{tilePoints}[i] = p$ B4: $\text{pointToIndex}[p] = (\text{int}) i$

stl in api 里面,

评价为代码写好了.



Name:

4

Problem 4 (10 points): Disjoint sets and UpTree

1. 2

2. 4

3.

Index	0	1	2	3	4	5	6	7	8	9
Arr	4	0	4	4	-7	4	3	-5	7	7

4.

Index	0	1	2	3	4	5	6	7	8	9
Arr	4	0	4	2	-10	4	3	4	7	7

5. $O(m \cdot \alpha(n))$ or $O(m \cdot \log^*(n))$
an approximation of $\alpha(n)$ is $\log^*(n)$

